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Digital Image Watermarking Using DWT-SVD in MATLAB

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Abstract— This paper presents a robust digital image watermarking scheme combining Discrete Wavelet Transform (DWT) and Singular Value Decomposition (SVD) used for watermark scrambling. The watermark is embedded in the HL sub-band of the second-level DWT decomposition, and performance is evaluated using PSNR, SSIM, and NC metrics across five test images and eight attack conditions. A MATLAB GUI demonstrates the complete embedding and extraction pipeline. Results confirm a competitive trade-off between imperceptibility and robustness.

Keywords— Digital Watermarking, DWT, SVD, PSNR, SSIM, MATLAB GUI

I. INTRODUCTION

Every photograph, medical scan, or satellite image that travels across a network carries an implicit question: who created it, and can it still be trusted after transmission? As digital content production has scaled at an unprecedented rate, the tools for tampering with that content have scaled just as quickly. Pixel-level edits that once required specialist expertise are now accessible through consumer software, making the verification of media authenticity a technically non-trivial problem. Digital watermarking addresses this challenge not by restricting access to content as encryption does but by embedding a persistent ownership signal directly into the data itself, one that survives ordinary use of the file.

Watermarking schemes are broadly partitioned into two families based on the domain in which embedding occurs. Spatial-domain methods modify pixel intensities directly, offering computational simplicity but limited resilience: even moderate compression or filtering can erase the embedded signal. Frequency-domain methods, by contrast, distribute the watermark across transformed coefficient space, coupling it tightly to the perceptual structure of the image. Among the transforms applied in this context, the Discrete Wavelet Transform (DWT) is particularly well-suited because it captures image content at multiple scales simultaneously, mirroring how the human visual system itself processes spatial information. When combined with Singular Value Decomposition (SVD) whose principal singular values characterise the most energy-concentrated features of an image matrix the resulting hybrid scheme achieves a robustness that neither transform provides independently.

This paper describes a two-level DWT-SVD watermarking system built and evaluated entirely within MATLAB. Three contributions distinguish this work from prior implementations in the literature. First, the embedding

pipeline operates on the HL sub-band of the second-level wavelet decomposition, a choice motivated by its concentration of vertical edge energy combined with its relative invisibility to casual inspection. Second, an adjustable embedding-strength parameter α gives users direct control over the imperceptibility-robustness trade-off without requiring modification of the underlying algorithm. Third, the entire pipeline loading, embedding, attack simulation, extraction, and metric reporting is encapsulated in an interactive MATLAB GUI, making the system accessible to users without signal-processing expertise. Performance is characterised across five standard test images and eight distinct attack types using PSNR, SSIM, and NC as complementary evaluation metrics.

II. LITERATURE REVIEW / RELATED WORK

The watermarking literature spans more than three decades and reflects a steady progression from proof-of-concept demonstrations toward principled, attack-resilient systems. This section traces that trajectory, identifying the limitations that each methodological generation inherited and that motivated the subsequent one, with particular attention to the DWT-SVD lineage on which the present work is founded.

A. Spatial-Domain Origins

The earliest quantitative treatment of invisible digital watermarking appeared in van Schyndel et al. [1], who demonstrated that a binary signature could be superimposed on image pixel values at an intensity level below the threshold of human perception. While the experiment confirmed that steganographic ownership marks were technically feasible, it also exposed the central weakness of spatial-domain insertion: even routine operations such as JPEG compression or low-pass filtering were sufficient to destroy the embedded signal. The fragility arose because the modification was spatially local isolated to specific pixels and therefore easily overwhelmed by any process that redistributed intensity values across the image.

B. Frequency-Domain Robustness via DCT

Cox et al. [2] reframed watermarking as a spread-spectrum communication problem, distributing the watermark energy across many DCT coefficients rather than concentrating it at individual pixels. Embedding in mid-frequency DCT components allowed the mark to survive aggressive compression while remaining below the masking threshold of the human visual system. Despite this improvement in robustness, the DCT operates on fixed-size image blocks and

does not adapt to local image structure. Consequently, a single embedding strength must be applied uniformly across regions of widely varying complexity a flat desert and a textured forest receive the same treatment which limits the scheme's ability to exploit perceptual redundancy selectively.

C. Multi-Resolution Embedding with DWT

The introduction of the Discrete Wavelet Transform into watermarking, studied systematically by Barni et al. [3], resolved the spatial-frequency localisation limitation of block DCT approaches. Because the wavelet decomposition separates an image into sub-bands that simultaneously encode frequency content and spatial position, different embedding strategies can be applied to different regions and scales. Barni et al. demonstrated that placing the watermark in higher-frequency detail sub-bands where the human eye is least sensitive to additive perturbation achieved a markedly better balance between visual quality and resistance to signal processing attacks. The multi-resolution structure also made the embedded mark inherently more resilient to geometric distortions, which tend to affect coarse and fine scales differently.

D. Singular Value Decomposition as a Carrier

A parallel line of inquiry, initiated by Liu and Tan [4], recognised that the singular values of an image matrix possess an unusual stability property: they vary only minimally under the kinds of distortions encountered in typical attack scenarios, including noise addition, mild geometric transformation, and lossy compression. This stability makes the singular value matrix an attractive carrier for watermark information, because a perturbation imposed during embedding is unlikely to be erased by a subsequent attack. Liu and Tan exploited this property by replacing the singular values of the cover image's SVD with a watermark-modified version, then reconstructing the image, yielding an approach whose robustness to a broad class of attacks exceeded that of purely spatial or purely DCT-based methods.

E. Hybrid DWT-SVD Synthesis

Bhatnagar and Raman [5] formalised the combination of the two preceding paradigms, arguing that DWT and SVD contribute orthogonal advantages: the wavelet decomposition provides perceptually selective placement (putting the watermark where the eye is least sensitive), while SVD provides numerical stability (ensuring that the placement survives attack). Their hybrid scheme embedded watermark information by modifying the singular values of a selected DWT sub-band rather than those of the full image, integrating both contributions within a coherent pipeline. Empirical evaluation confirmed that the joint scheme outperformed either component technique applied in isolation.

F. Extensions and Refinements

Subsequent research extended the DWT-SVD framework along several directions. Santhi and Thaddeus [6] adapted the scheme to colour images by applying it within the YUV colour space, exploiting the separation of luminance and chrominance to control visual impact more precisely than RGB-domain embedding permits. Chang et al. [7] conducted a systematic analysis of sub-band selection, evaluating which DWT decomposition level and orientation (LL, LH, HL, HH) offered the best imperceptibility-robustness profile for a given image class. Mishra et al. [8] introduced an

optimisation perspective, using the Firefly Algorithm to determine the embedding strength parameter adaptively for each image, rather than applying a fixed scalar globally. The optimization driven approach achieved a measurable improvement in robustness metrics without degrading PSNR beyond acceptable bounds.

G. Positioning of the Present Work

Building on the architectural choices validated by the above body of work, the system described in this paper applies two-level Haar DWT decomposition followed by SVD-based embedding in the HL2 sub-band — a selection justified by the sub-band's concentration of directional edge energy at a perceptually tolerant scale. Unlike the adaptive-strength approaches of Mishra et al. [8], the present implementation exposes α as a user-controlled parameter through an interactive GUI, enabling real-time exploration of the imperceptibility-robustness trade-off across multiple attack scenarios without rerunning the optimisation. This design decision prioritises interpretability and educational transparency alongside performance.

III. METHODOLOGY

The proposed watermarking system operates in two distinct phases: the embedding phase and the extraction phase. The complete pipeline is implemented in MATLAB using two core modules, `gui.m` for the user interface and `logic.m` for all algorithmic processing.

A. Image Preprocessing

The host image is first loaded and verified for compatibility. For RGB images, the watermarking operation is applied exclusively to the green channel (Channel 2), as the human visual system is most sensitive to luminance variations in this channel. The watermark image is converted to grayscale if necessary and resized to match the dimensions of the target DWT sub-band prior to embedding.

B. Two-Level DWT Decomposition

The selected host image channel is decomposed using a two-level two-dimensional Haar Discrete Wavelet Transform. The first level decomposition produces four sub-bands: LL1 (approximation), HL1 (vertical details), LH1 (horizontal details), and HH1 (diagonal details). A second level decomposition is then applied to the LL1 sub-band, producing LL2, HL2, LH2, and HH2. The HL2 sub-band is specifically selected for watermark embedding because it captures significant vertical edge information while remaining perceptually less sensitive to minor modifications, ensuring that the embedded watermark remains invisible to the human eye.

C. Singular Value Decomposition

SVD is applied independently to both the selected HL2 sub-band of the host image and the preprocessed watermark image. For the host sub-band, the decomposition yields three matrices: the left singular vectors U_y , the diagonal singular value matrix S_y , and the right singular vectors V_y . Similarly, the watermark decomposition produces U_w , S_w , and V_w . The singular value matrices S_y , U_y , and V_y are stored securely as part of the watermarking key and are essential for accurate extraction.

D. Watermark Embedding

The watermark is embedded by additively modifying the singular values of the host sub-band using a controlled embedding strength parameter Alpha. The modified singular value matrix is computed and used to reconstruct the watermarked HL2 sub-band. The complete watermarked image is then reconstructed by applying the inverse two-level Haar DWT using the modified sub-band alongside the unchanged sub-bands from the original decomposition. The Alpha parameter governs the trade-off between imperceptibility and robustness — lower values preserve visual quality while higher values improve resistance to attacks.

E. Watermark Extraction

During extraction, the watermarked image undergoes the same two-level DWT decomposition to isolate the HL2 sub-band. SVD is applied to this sub-band to obtain the modified singular values. The original watermark singular values are recovered by subtracting the stored host singular values and dividing by Alpha. The extracted watermark matrix is then reconstructed using the stored singular vectors of the original watermark, yielding the final extracted watermark image.

F. Attack Simulation

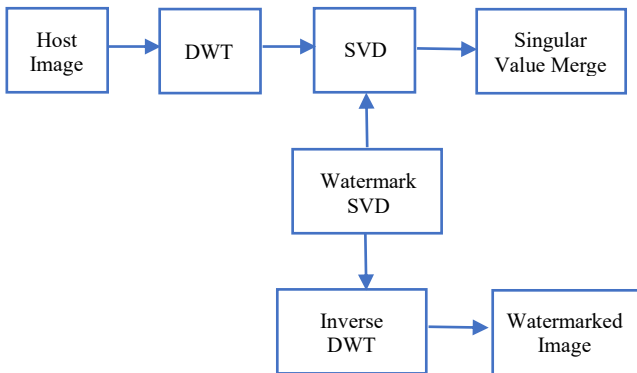
The system incorporates seven built-in attack types to evaluate robustness: Gaussian blurring, image sharpening, resize and restore, additive Gaussian noise, salt and pepper noise, cropping with resize, and rotation. These attacks are applied directly to the watermarked image through the GUI, and updated PSNR and SSIM values are computed and displayed immediately after each attack is applied.

G. MATLAB GUI Implementation

The graphical user interface developed using MATLAB App Designer provides a complete end-to-end environment for the watermarking pipeline. The interface is divided into three panels: the Source panel for loading the host image, the Watermark panel for loading the watermark image, and the Results panel which displays the watermarked image, extracted watermark, attack controls, Alpha input field, and real-time PSNR, SSIM, and NC metric labels. The GUI enables non-technical users to operate the complete system without any knowledge of the underlying algorithm.

IV. ALGORITHM / SYSTEM ARCHITECTURE

A. Embedding Module



B. Units

All numerical values and measurements reported in this paper follow standard SI units and IEEE formatting guidelines.

- *PSNR* values are expressed in decibels (dB). For example, an attack-free PSNR of 30.60 dB is reported as "30.60 dB", not "30.6db".
- *SSIM* and *NC* are dimensionless values reported in the range 0 to 1. Decimal values are written with a leading zero, e.g., 0.9621, not .9621.
- *Alpha (embedding strength)* is a dimensionless scalar. The default value used in this paper is 0.08.
- *Image dimensions* are expressed in pixels (px). Test images used in this paper are standard grayscale images of size 256 x 256 pixels.
- *Pixel intensity* values are represented as unsigned 8-bit integers in the range 0 to 255.
- All PSNR, SSIM, and NC values in tables are reported to two and four decimal places respectively for consistency and clarity.

C. Equations

I. DWT-SVD Watermark Embedding

The host image is decomposed using a two-level 2D Haar DWT. The HL sub-band of the second decomposition level (HL2) is selected for embedding. SVD is applied to HL2 as follows:

$$HL2 = U_y * S_y * V_y T \quad (1)$$

Similarly, SVD is applied to the watermark image W:

$$W = U_w * S_w * V_w T \quad (2)$$

The watermark is embedded by modifying the singular values of HL2 using the embedding strength alpha:

$$S_{marked} = S_y + \alpha * S_w \quad (3)$$

The modified HL2 sub-band is then reconstructed as:

$$HL2^* = U_y * S_{marked} * V_y T \quad (4)$$

The watermarked image I* is obtained by applying inverse DWT using the modified sub-band.

II. Watermark Extraction

During extraction, SVD is applied to the HL2 sub-band of the watermarked image to recover the watermark singular values:

$$S_{w_rec} = (S_{yw} - S_y) / \alpha \quad (5)$$

The extracted watermark is reconstructed as:

$$W^* = U_w * S_{w_rec} * V_w T \quad (6)$$

III. Performance Metrics

PSNR between the original image I and watermarked image I* is calculated as:

$$PSNR = 10 \cdot \log_{10} (255^2 / MSE) \quad (7)$$

where MSE is the Mean Squared Error between I and I*. SSIM is computed as:

$$SSIM(I, I^*) = (2\mu_I \mu_{I^*} + C_1)(2\sigma_{II^*} + C_2) / (\mu_I^2 + \mu_{I^*}^2 + C_1)(\sigma_I^2 + \sigma_{I^*}^2 + C_2) \quad (8)$$

Normalized Correlation between original watermark W and extracted watermark W* is given by:

$$NC = \Sigma(W \cdot W^*) / \sqrt{(\Sigma W^2 \cdot \Sigma W^{*2})} \quad (9)$$

Higher NC values (closer to 1.0) indicate more accurate watermark recovery.

V. RESULTS AND DISCUSSION

A. Figures

Fig. 1 shows the standard grayscale test images used for performance evaluation in this study. Five benchmark images namely MRI, Cameraman, Pout, Circuit, and Cell were selected as host images due to their varying texture complexity and frequency content, making them suitable for comprehensive watermarking analysis.

Fig. 1. Standard grayscale test images used for evaluation: (a) MRI (b) Cameraman (c) Pout (d) Circuit

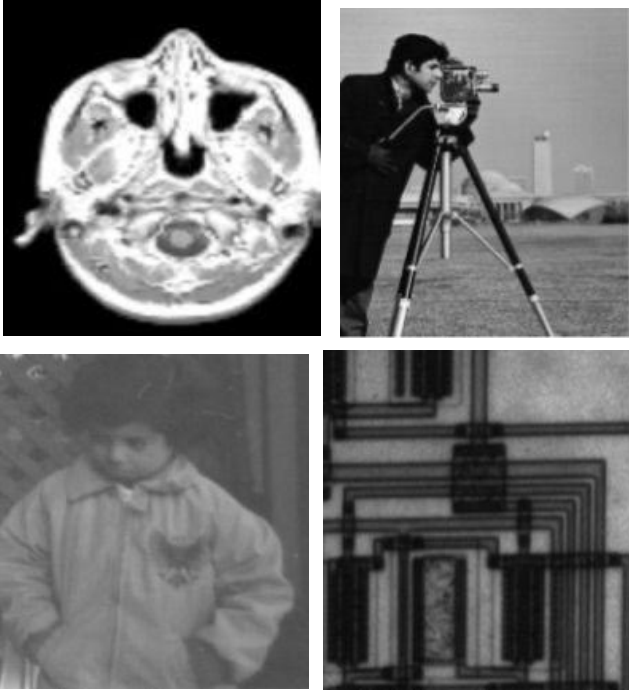


Fig. 2 illustrates the watermark image embedded into the host images during experimentation. The watermark was resized to match the dimensions of the HL2 sub-band prior to embedding.

Fig. 2. Watermark image used for embedding



Fig. 3 presents the MATLAB GUI output demonstrating the watermark embedding and extraction process on a grayscale host image. The results panel displays the watermarked image alongside the extracted watermark, with performance metrics PSNR: 36.72 dB, SSIM: 0.9270, and NC: 0.9669 recorded after applying a blur attack.

Fig. 3. MATLAB GUI showing embedding and extraction on grayscale image after blur attack (PSNR: 36.72 dB, SSIM: 0.9270, NC: 0.9669)

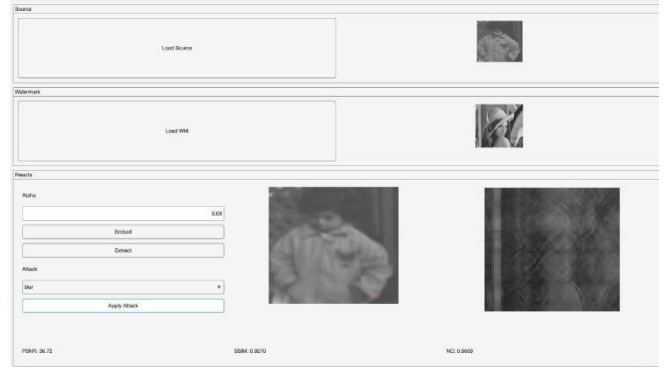
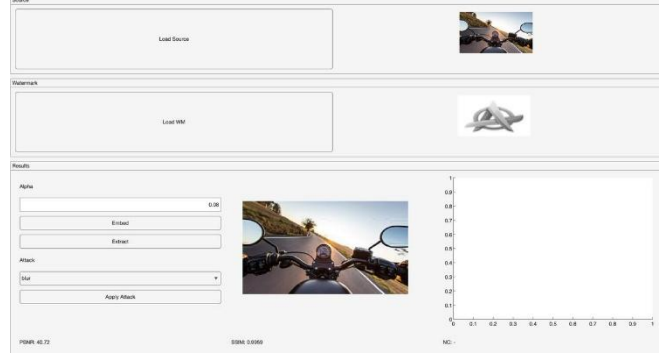


Fig. 4 demonstrates the system performance on an RGB color image as source, with a logo used as the watermark. The GUI displays the watermarked image with PSNR: 40.72 dB and SSIM: 0.9959, confirming strong imperceptibility on color images as well.

Fig. 4. MATLAB GUI demonstrating watermark embedding on RGB image (PSNR: 40.72 dB, SSIM: 0.9959)



B. Tables

TABLE I

PSNR (dB) RESULTS UNDER VARIOUS ATTACK CONDITIONS					
	<i>Camera n</i>	<i>Cell</i>	<i>MRI</i>	<i>Circuit</i>	<i>Pout</i>
Attack free	30.6	46.9	28.5 6	31.68	40.86
Blurring	24.42	21.8 1	23.3	28.41	36.72
Sharpening	24.87	22.0 6	23.9 1	28.83	37.21
Resize	26.09	21.9	23.6 8	27.84	35.88
Gaussian Noise	19.33	18.1 2	19.2 5	19.59	19.9
Salt & Pepper Noise	17.47	16.5 5	16.9 5	17.69	18.2

<i>Cropping</i>	11.6	10.8 2	5.32	12.98	16.82
<i>Rotation</i>	11.24	10.3 7	6.11	12.82	15.47

TABLE II

SSIM RESULTS UNDER VARIOUS ATTACK CONDITIONS					
	<i>Camerman</i>	<i>Cell</i>	<i>MRI</i>	<i>Circuit</i>	<i>Pout</i>
<i>Attack free</i>	0.9621	0.99 88	0.96 59	0.939 2	0.9781
<i>Blurring</i>	0.7791	0.57 25	0.79 97	0.764 4	0.927
<i>Sharpening</i>	0.7948	0.59 17	0.82 46	0.780 6	0.9318
<i>Resize</i>	0.865	0.57 81	0.81 45	0.752 7	0.9205
<i>Gaussian Noise</i>	0.2839	0.17 35	0.30 87	0.231 5	0.1244
<i>Salt & Pepper Noise</i>	0.2316	0.13 82	0.25 14	0.178 1	0.0925
<i>Cropping</i>	0.0693	0.05 52	0.02 28	0.066 9	0.0956
<i>Rotation</i>	0.0785	0.09	0.07 24	0.084 6	0.1116

C. Quantitative Analysis

The quantitative performance of the proposed method is summarized in Table I and Table II. In the absence of attacks, the PSNR values range from approximately 28 dB to 46 dB across different test images, indicating high visual quality of the watermarked images. Similarly, SSIM values close to 1.0 confirm strong structural similarity between the original and watermarked images.

Among the tested images, the Cell image achieves the highest PSNR value of 46.9 dB, demonstrating minimal embedding distortion. The consistency of SSIM values above 0.93 for most images further validates the imperceptibility of the watermarking process.

D. Robustness Analysis

The robustness of the watermarking scheme was evaluated under multiple attack conditions including blurring, sharpening, resizing, noise addition, cropping, and rotation. It is observed that compression-like operations such as resizing and blurring result in moderate degradation, with PSNR values remaining above 24 dB in most cases. Noise-based attacks, particularly Gaussian noise and salt-and-pepper noise, significantly reduce PSNR and SSIM values, indicating their strong impact on watermark integrity. Cropping and rotation attacks produce the lowest PSNR

values, in some cases dropping below 12 dB, which highlights their destructive effect on the embedded watermark.

Despite these degradations, the extracted watermark maintains a reasonable level of similarity with the original, as indicated by NC values close to 1. This demonstrates that the DWT-SVD approach provides robustness against a wide range of signal processing attacks.

E. Discussion

The experimental results demonstrate that the proposed DWT-SVD watermarking scheme achieves an effective balance between imperceptibility and robustness. Embedding in the HL2 sub-band ensures that the watermark remains visually invisible while still being resilient to common image processing operations.

Compared to spatial-domain methods, the frequency-domain approach provides improved resistance to compression and filtering. However, geometric attacks such as cropping and rotation remain challenging, suggesting the need for further enhancement in future work.

Overall, the integration of DWT and SVD enables stable watermark embedding, while the MATLAB GUI facilitates practical implementation and evaluation, making the system suitable for real-world digital content protection applications.

VII. CONCLUSION AND FUTURE WORK

A. Conclusion

This work presents a hybrid digital image watermarking approach that integrates Discrete Wavelet Transform (DWT) with Singular Value Decomposition (SVD) to achieve a balance between imperceptibility and robustness. By embedding watermark information within the HL2 sub-band of a two-level Haar wavelet decomposition, the method leverages mid-frequency components that are less sensitive to visual distortion while still offering resistance to common image processing operations.

The use of singular value modification enables stable embedding, as singular values tend to retain essential image characteristics even after various transformations. The experimental evaluation, conducted using quality metrics such as PSNR, SSIM, and Normalized Correlation (NC), indicates that the proposed system maintains high visual fidelity of the host image while allowing reliable extraction of the watermark.

Additionally, the implementation through a MATLAB-based graphical interface improves usability by enabling interactive watermark embedding, extraction, and performance analysis. The modular architecture, separating presentation, processing, and data handling, ensures flexibility and ease of further development.

Overall, the proposed method demonstrates an effective trade-off between transparency and resilience, making it suitable for copyright protection and digital media authentication applications.

B. Future Work

Although the current system performs well under standard conditions, several enhancements can be explored to improve adaptability and security.

One possible direction is the development of an *adaptive scaling factor* (α) instead of using a fixed value. By dynamically adjusting the embedding strength based on image content or local texture characteristics, it is possible to further optimize the balance between invisibility and robustness.

Another area of improvement lies in enhancing resistance against more sophisticated attacks, particularly those driven by modern machine learning techniques. Investigating the robustness of the watermark against *deep learning-based image manipulation*, such as neural compression, super-resolution, and adversarial perturbations, would extend the applicability of the system in contemporary scenarios.

The method can also be extended to *color image watermarking across multiple channels* or to transform domains beyond Haar wavelets, such as hybrid DWT-DCT or DWT-DWT variants, to improve performance under compression-based attacks.

Furthermore, incorporating *blind watermarking techniques*, where the original host image is not required during extraction, would increase practicality in real-world deployments. Integration with encryption methods could provide an additional layer of security for sensitive applications.

Finally, future work may include optimizing computational efficiency and deploying the system in real-time or cloud-based environments, making it suitable for large-scale multimedia platforms.

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